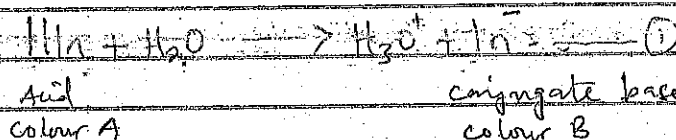


Acid-Base Titration

An acid-base titration is a laboratory procedure used in quantitative chemical analysis to determine the concentration of either a base or an acid. It is often carried out using a visual indicator to determine the end-point. An acid of known concentration is added from a burette to an alkali solution (base) of unknown concentration in a conical flask. The end point occurs when stoichiometric amount of the acid has been added to the base. At this point complete neutralization is achieved. The solution consists of salt and water only. From the values of the molarity of the acid, the volume of acid used and the volume of base used, the molarity of the base can be determined.

Indicators {Acid-Base}

An acid-base indicator is a laboratory indicator substance which has one colour in an acid & another colour in a base. They are usually weak acids or bases. Acid-base indicators are also known as pH indicators. Consider an indicator which is a weak acid with general formula HIn . When they dissolved in water, an equilibrium is established between the weak acid & its conjugate base.



We can apply the equilibrium law to the above expression. In general, for a weak acid HIn , the dissociation constant is given by:

$$K_{In} = \frac{[H_3O^+][In^-]}{[HIn]} \quad (2)$$

where K_{In} is called the Indicator Dissociation Constant. The colour of an indicator changes from A to B or vice-versa at a point called the turning point, at this point the concentration of the acid HIn is equal to the concentration of the conjugate base In^- .